

IAB330_2a_Group23

Design Proposal



Online Workshop Students

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20/9/2024

1) Introduction

The prototype Machine Learning (ML)-based Human Activity Recognition (HAR) system is designed to classify various human activities using sensor data collected from a wearable device. HAR systems are particularly useful in fitness tracking and health monitoring, providing valuable insights into activity patterns and overall health (Bulling et al., 2014). The specific aim of this project is to develop a fitness tracker capable of recognising different types of workout activities, including walking, climbing stairs, sitting and jumping, using an ADXL343 accelerometer sensor attached to the user.

The scope of this project includes the integration of the accelerometer with a Raspberry Pi to capture, process and classify movement data in real time. By leveraging machine learning techniques, the system will classify activities based on the sensor data, focusing primarily on detecting walking, jumping, sitting and climbing stairs. The project does not involve features like tracking heart rate, calories burned or sleep patterns, as the focus is on recognising physical movements using accelerometer data (Lane et al., 2015). This choice ensures that the system is optimised for specific workout activities without adding unnecessary complexity.

The project's relevance is within the context of the growing use of wearable technology for fitness and health applications. With an increasing number of users relying on wearable devices for activity tracking and personalised health insights, a system capable of accurately distinguishing between workout activities will provide significant value, enabling users to track their fitness progress and receive real-time feedback (Ravi et al., 2005). By focusing on core workout activities, this prototype offers a targeted approach to fitness tracking, making it suitable for both casual users and athletes.

Fitness Tracker for Specific Workouts:

Concept: Develop a wearable device that can recognize different types of workout activities, such as running, cycling, squats, and push-ups.

Sensors: Accelerometer (e.g., ADXL343)

Use Case: Provides feedback on workout intensity and tracks repetitions for strength training exercises.

Challenges: Accurately differentiating between similar movements (e.g., squats vs. lunges).

2) Design Objectives

Project Scope:

The project's purpose is to design and build a device for recognizing human activity, one that employs the techniques of machine learning, and whose main component is an ADXL343 accelerometer. The device is being built around a Raspberry Pi, which will handle the system's computation and connectivity tasks. The idea is to detect and classify four key activities: walking, sitting or standing still, jumping, and climbing stairs. These activities have been selected for their relevance to real-world applications—knowing when a person is performing any of these four acts could be quite valuable across a range of contexts—and because they correspond to what the ADXL343 does best, which is recognising accelerative events along three axes (X, Y, Z), making it well-suited for monitoring a broad range of human movements (Mathie, 2004)

This project does not cover the monitoring of heart rate, calories burned, steps taken, or sleep patterns, which are outside the sensor's designed capabilities and our project's focus. Instead, the primary aim of this project is to create a prototype of a wearable activity tracker that can tell what kind of activity a user is doing; for instance, the tracker should be able to discern between walking, climbing stairs, and being totally inactive.

Objectives:

The primary objective of this project is to create a wearable fitness tracker capable of accurately detecting a variety of activities, such as walking, jumping, climbing stairs, and periods of inactivity.

The device will detect the following key activities:

Walking: As a common daily activity, walking produces a distinct oscillating acceleration pattern that the ADXL343 sensor can easily detect. Walking is essential for assessing general mobility and activity levels, which are important health monitoring, fitness tracking, and rehabilitation metrics. Its rhythmic nature aligns well with the sensor's ability to capture repetitive movement patterns.

Sitting/Standing Still: Recognising periods of inactivity is crucial for monitoring sedentary behaviour, which is important for overall health. The ADXL343 sensor's sensitivity to even slight movements allow it to effectively differentiate between sitting, standing still, and low-movement activities. It is ideal for applications that emphasise rest periods or inactivity.

Jumping: Jumping involves rapid and pronounced vertical acceleration changes, particularly along the Z-axis, which the ADXL343 sensor can capture accurately. By

identifying the sharp spikes in acceleration during takeoff and landing, the system can reliably detect this high-intensity activity, making it a valuable feature for fitness tracking and exercise monitoring.

Climbing Stairs: This activity combines vertical motion with walking. The ADXL343 sensor's capability to track changes along the Z-axis enables the system to distinguish between walking on flat surfaces and climbing stairs, providing more detailed insights into physical exertion and energy use.

3) System Specification

This section provides an in-depth overview of the prototype for the (HAR) system, with the emphasis being on the hardware configuration, software and communication protocols. The centrepiece of the system is the ADXL343 accelerometer, and the data that it collects are ultimately processed using a Raspberry Pi (Choudhury, 2024). The design choices made for the prototype particularly those related to the configuration of the system's key components, the handling of data, and the management of power and errors are discussed.

Sensor Configuration

The prototype will be placed most conveniently for the user which is typically at there waist. This has been determined as the best placement for the device to track a user during a variety of movements associated with different activities. The collected data will be the most accurate if the sensor maintains the orientation for which it was designed with the Z-axis aligned vertically (Dam, 2024). The data will assist in understanding the correlation between which kinds of movements the user is making while wearing the device and what kinds of actions they are performing in the real world. (Dam, 2024).

Operating range

The (HAR) system is well suited for the $\pm 4g$ range. When applied to an accelerometer, this range ensures the detection of a movement spectrum that encompasses both the weak accelerations associated with human actions (e.g., a person standing or sitting) and the intense ones generated by fast movements (e.g., a person jumping). Research has shown that many commonly performed human movements fall between these two thresholds (Stevens, 2021).

Additionally, this range maintains a good level of resolution, allowing for precise detection of smaller movements, while preventing excessive noise that might occur with higher sensitivity settings. Therefore, the 4g range strikes an optimal balance, enabling accurate and efficient activity recognition across a variety of motion intensities (Stevens, 2021).

Sampling frequency

To minimize the noise in the results from the accelerometer, a sampling rate of 20 Hz will be used. This selection is based on the types of human activities that have been selected, which are generally well below 5 Hz in frequency (Kitch, 2023). Experiments have revealed that sampling using a higher rate not only results in more data to process but also may capture high-frequency "noise" that is actually irrelevant to the activity being assessed, resulting in less accurate and efficient activity recognition (Kitch, 2023).

Based on the Nyquist theorem, a 20hz sampling rate is sufficient to accurately capture human movements while maintaining data efficiency (Kitch, 2023). It strikes a balance between retaining meaningful motion data and preventing the system from being overwhelmed with unnecessary information, resulting in more accurate and efficient activity recognition (Kitch, 2023).

In summary, 20 Hz is a good choice for reducing noise in HAR systems because it focuses on capturing the lower-frequency components associated with human activities while filtering out irrelevant high-frequency noise, leading to more accurate and efficient data processing (Kitch, 2023).

Software and Communication Protocols

The software is responsible for managing the data flow from the sensor to the processing unit, executing algorithms for activity recognition, and handling communication between components.

Data Collection: Raw acceleration data will be collected by the Raspberry Pi from the ADXL343 sensor through the I²C communication protocol. The "Inter-Integrated Circuit" protocol is a popular choice in embedded systems because of its simplicity and efficiency. Because only two wires are needed (SDA for data and SCL for the clock), you can use it to communicate with multiple devices on the same bus (Campbell, 2016). Each device can be uniquely addressed, which makes I²C a preferred method for reducing wiring complexity. Unlike SPI, which needs multiple lines per device, I²C keeps the number of required GPIO pins to a minimum. That's an advantage in systems where you need to conserve GPIO pins (Campbell, 2016).

Additionally, its widespread support across various devices, ease of use, and pin efficiency make I²C a versatile and suitable choice for systems like (HAR) wearable devices, where multiple sensors, such as accelerometers, need to be integrated with minimal wiring and hardware complexity (Campbell, 2016).

Data Processing: The Python library `adafruit_adxl34x` will be used for I²C communication with the accelerometer, `numpy`, `pickle`, and `pandas` for data manipulation. These libraries enable the Raspberry Pi to read raw data from the ADXL343 and to preprocess it, calculating key features, like the mean acceleration, correlation between axes, and the range of motion (Clark, 2019). This is then fed into a machine learning (ML) model trained to recognize activities like walking, sitting, and jumping, etc. The ML model uses the popular library `scikit-learn` for device-level inference, allowing it to classify activities in real time. Meanwhile, I²C is a crucial part of

the hardware communication layer, ensuring that activity recognition happens efficiently and responsively. (Saha et. al, 2022).

Data Noise: The system is designed to filter most of the irrelevant data and noise occurring through the configuration of the system (Liu, 2024). However, if it is deemed that the system is experiencing any excess noise a noise filtering program to remove any noisy data caused by sensor jitter or minor, unintentional movements (e.g., vibrations) can be implemented. This filtering ensures a smooth recording of the activity. (Liu, 2024)

Power Source

The prototype is powered by a rechargeable lithium-polymer (Li-Po) battery, chosen for its high energy density, lightweight design, and ability to provide stable power for extended use. The battery capacity is carefully selected to ensure continuous operation. Its durability and lightweight nature make the Li-Po battery ideal for a wearable system (Yurur et al., 2016).

Error Handling

A system's reliable and robust operation is ensured by good error handling. Continuous and uninterrupted operation is possible even when there are hardware or software malfunctions. Adverse conditions like failed sensors or broken communications do not take down a well-designed system. The mechanism that provides a faulty system with the appearance of a functional one is the most basic layer of error handling (Palaniswami et al., 2013).

Loose connections and misconfigured addresses can lead to communication errors between the Raspberry Pi and the ADXL343 sensor. Should such an error occur, the software can attempt to remedy the situation by retrying the command that failed. If the command still times out after a few retries, the system is notified so that the user may investigate the problem. A more sophisticated way of dealing with intermittent issues is to periodically scan the I²C bus and check for the presence of all expected devices. If a device is not found, that is a clear indication of a problem. (Palaniswami et al., 2013).

Sometimes, the accelerometer produces invalid data. This can happen because of noise, power problems, or transient hardware issues. For example, accelerometer readings might show a sudden spike or extremely low reading that just doesn't make sense for a human in motion. We can check the validity of accelerometer data by simply comparing it against a predetermined reasonable range. If we know the expected range of human movement, we can flag anything that drastically exceeds those physical limits

as suspect. A common signal processing technique for smoothed out noisy data is a low-pass filter.

The Raspberry Pi may abruptly stop when a sudden power cut or power failure occurs, which can disrupt data collection and lead to a loss of information and possible system corruption. This scenario can be avoided by saving the sensor data and the state of the system automatically and at regular intervals—that is, not waiting until the end of a session to save. Because of the nature of a power cut, there's no time to gracefully exit, so saving and reducing activity in a just-in-case mode can prevent the loss of significant data and give the system a little insurance against corruption during an ungraceful exit.

Sampling rates that are not consistent can happen because of system latency or from outside forces that impinge on the processing capability of the Raspberry Pi. Latency—whether it's inherent in the system or caused by external forces—can affect the speed of data collection, which means the sampling rate can become "slushy." Conversely, if the sampling rate is too fast, data can get jammed up in "buffers" and processed in bursts when the system catches up. That's not a good way to accomplish fast, real-time processing. So, how do we solve the problem? First, a system reference clock is used to timestamp data collected from the three accelerometers. This ensures that the data is in a synchronized state and that we're measuring accurately three-dimensional (3D) coordinates. (Palaniswami et al., 2013).

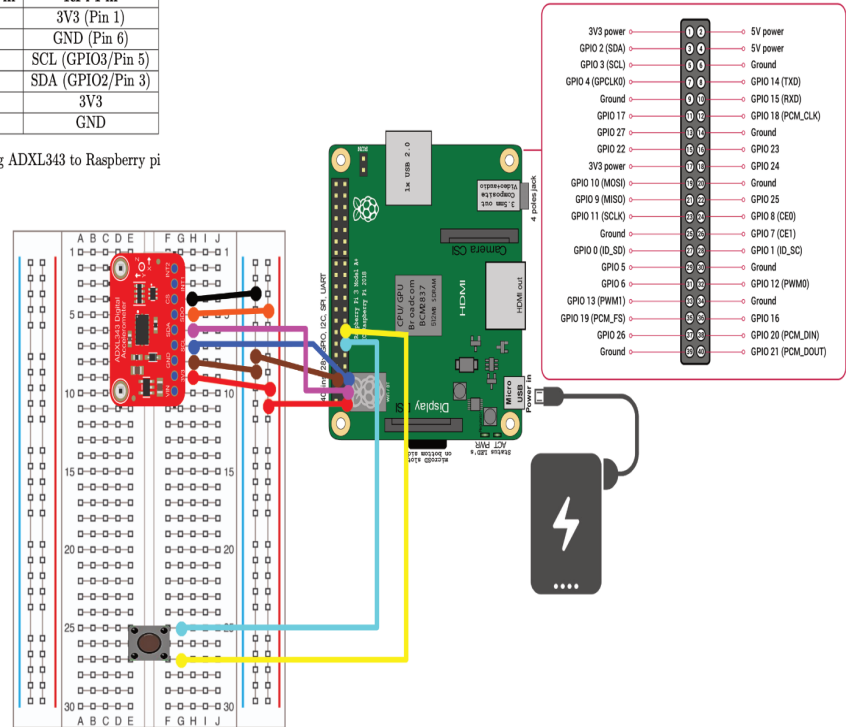
Software bugs, unhandled exceptions, and hardware problems can cause unpredictable failures. To cope with this, the software should catch and handle errors that might occur during critical operations. It is especially important to do this during I²C communication, reading data, and making predictions with the model since errors in these parts can lead to serious malfunctions. The system can log error messages to give the user a clue about where the problem is. It can also send alerts through various means if the user really needs to know about the error right away.

To sum up, a HAR system uses preventive, reactive, and robustness strategies to ensure it maintains dependable performance and a correct recognition of activities (Palaniswami et al., 2013). The system can handle a mix of errors—from wiry, uncooperative humans to unreliable communications—without having to rigorously always check over all a person's activities. These measures work together to keep the system working under a variety of conditions, reliably recognizing the activities of the user across a virtual environment.

System Diagram

ADXL343 Pin	RPi Pin
VCC	3V3 (Pin 1)
GND	GND (Pin 6)
SCL	SCL (GPIO3/Pin 5)
SDA	SDA (GPIO2/Pin 3)
CS	3V3
SDO	GND

Table 1: Wiring ADXL343 to Raspberry pi



4) ML Model Development

In the development of our Human Activity Recognition (HAR) system, we have selected the Decision Tree Classifier And the Random Forest Classifier as our primary machine learning models. These models are recognised for their efficacy in classification tasks, particularly with structured sensor data such as accelerometer readings (Breiman, 2001; Kuhn & Johnson, 2013).

Decision Tree Classifier:

The Decision Tree classifier operates by learning decision rules derived from data features, systematically splitting the dataset into classes (Quinlan, 1993). Its inherent interpretability and ability to capture non-linear relationships make it an appropriate choice for initial explorations of our HAR system.

Key Configurations:

- **Maximum Depth:** The model's complexity will be controlled by experimenting with different maximum depths to mitigate overfitting.
- **Criterion:** The quality of splits will be measured using either the 'gini' impurity or 'entropy', with the final choice being guided by empirical performance during cross-validation.

Rationale: The Decision Tree classifier's simplicity enables a foundational understanding of the dataset's structure and feature importance. While its propensity to overfit can be a limitation, this model serves as a valuable baseline for evaluating the feasibility of activity recognition based solely on accelerometer data (Breiman, 2001; Kuhn & Johnson, 2013).

Expected Performance: Although the Decision Tree may exhibit overfitting tendencies, particularly in noisy sensor data scenarios, it is anticipated to provide an initial benchmark for recognizing patterns in human activities.

Random Forest Classifier:

Random Forests employ an ensemble learning approach by constructing multiple decision trees and aggregating their outputs to enhance classification accuracy and robustness (Breiman, 2001). This method is particularly suited for complex datasets, such as accelerometer signals, that encompass variability and overlapping characteristics among different activities.

Key Configurations:

- **Number of Trees (n_estimators):** We will initiate with an ensemble of 100 trees, providing a balance between robustness and computational efficiency.
- **Maximum Features:** Limiting the number of features considered for each split aims to reduce model variance, with optimal values determined through parameter tuning.

Rationale: Random Forests are capable of capturing intricate interactions between features, offering greater resilience against overfitting compared to a single decision tree. Their ensemble nature makes them well-suited for the diverse and complex motion patterns inherent in human activities.

Expected Performance: It is anticipated that Random Forests will outperform the Decision Tree classifier in terms of accuracy and generalisation, particularly when dealing with the noisy and overlapping nature of accelerometer data.

Feature Extraction

Feature extraction is a pivotal step in transforming raw accelerometer data into representative inputs for machine learning models. Given the unavailability of additional sensors such as a gyroscope, our approach emphasises deriving informative features from the accelerometer to distinguish effectively between various workout activities (Ravi et al., 2005; MathWorks, n.d.).

Time-Domain Features:

Mean, Variance, and Standard Deviation: These statistical measures are extracted from the acceleration data along each axis (X, Y, and Z). They encapsulate the intensity and variability of movements, facilitating the differentiation between high-intensity activities, such as running, and more controlled movements, like squats.

Signal Magnitude Area (SMA): The SMA is computed to aggregate the magnitude of accelerometer signals across all axes. This measure captures the overall body movement and helps differentiate between activities with varying movement amplitudes, such as the dynamic nature of running versus the localised movement in arm curls.

Zero-Crossing Rate: This feature quantifies the rate at which the acceleration signal changes direction, providing insight into repetitive motion patterns. It is particularly useful for identifying activities characterised by periodic motions, such as running or squatting.

Frequency-Domain Features:

Fast Fourier Transform (FFT) Coefficients: FFT is applied to analyse the frequency components of the accelerometer signals. Different activities exhibit distinct frequency patterns, enabling the identification of characteristic movements. For instance, the regularity of walking and the sporadic bursts of energy in jumping can be distinguished through their spectral energy and dominant frequency components.

Justification for Feature Selection:

The combination of time-domain and frequency-domain features offers a comprehensive representation of the accelerometer signals (Vinh et al., 2017). Time-domain features provide insights into the magnitude and variability of movements, while frequency-domain features capture the rhythmic and periodic nature of activities. This multi-faceted approach enhances the model's capability to discern between activities with varying motion dynamics, such as steady walking versus dynamic jumping jacks.

Evaluation

The evaluation of the HAR system is a critical component to ensure the accuracy and reliability of activity recognition. The following metrics have been selected for a comprehensive assessment of model performance:

Accuracy: This metric provides an overall measure of the model's correctness by computing the proportion of accurately classified activities to the total number of activities. While accuracy offers a general performance overview, it can be less informative in the presence of class imbalances.

Precision and Recall:

Precision: Precision measures the model's ability to correctly identify instances of a specific activity without erroneously classifying other activities as this one. High precision is particularly crucial for activities requiring precise identification, such as distinguishing squats from other leg movements.

Recall: Recall, or sensitivity, assesses the model's capability to identify all instances of a particular activity. High recall is essential in contexts such as real-time monitoring, where missing instances (e.g., missed falls) can have significant consequences.

F1-Score: The F1-score is the harmonic mean of precision and recall, providing a balanced evaluation metric, especially in scenarios with uneven class distributions (Pedregosa et al., 2011). It is particularly useful in this context, where certain exercises may be less represented in the dataset.

Confusion Matrix: A confusion matrix will be employed to visualise the classification performance across different activities. This analysis aids in identifying specific activity pairs that are frequently misclassified, guiding further model refinement.

Evaluation Plan:

Cross-Validation: To enhance the model's generalisation capability, k-fold cross-validation (e.g., 5-fold) will be implemented. This method ensures that the model's performance is consistent across varied data subsets, mitigating the risk of overfitting.

Train-Test Split: Initially, the dataset will be divided into training and testing sets (e.g., 70%-30%) to facilitate model training and independent evaluation. The test set serves as a final assessment of the model's predictive power on unseen data.

Through a careful selection of models, a strategic feature extraction process, and a rigorous evaluation framework, this section outlines a comprehensive machine learning strategy aimed at the accurate recognition of diverse workout activities using accelerometer data.

5. Testing and Evaluation

To ensure the functionality, accuracy, and reliability of the HAR system, a detailed testing plan is essential. This testing plan aims to comprehensively verify that each component operates correctly, and that the system can accurately classify activities in real-world scenarios. Both component-level and system-level testing will be performed to ensure consistent performance across different conditions.

Component-Level Testing

Accelerometer Calibration and Data Accuracy

The ADXL343 accelerometer will undergo calibration tests to ensure precise data capture across the X, Y and Z axes. This involves placing the sensor in controlled environments such as flat surfaces, vertical orientations, and other fixed positions to compare the recorded data against expected values. Given that accurate movement detection relies on the sensor's precision, any deviation will be corrected to minimise errors. This is crucial as inaccuracies in accelerometer data could lead to faulty activity classification (Almeida et al., 2015).

Data Transmission

The reliability of data transmission between the accelerometer and the Raspberry Pi will be evaluated by testing the I2C communication protocol. Wired transmission tests will measure the consistency and completeness of the data flow, ensuring that no packets are lost during transmission. Any transmission failures can directly affect the system's ability to classify activities correctly, leading to errors in real-time classification. The integrity of data transmission will be a key factor in maintaining the overall accuracy of the HAR system (Dobre & Xhafa, 2014).

Data Processing Pipeline

Each stage of the data processing pipeline, from data collection through feature extraction, will be tested. Known datasets will be used to verify that the processing functions, including segmentation, feature extraction and classification, are accurate and efficient. Real-time processing with low latency is crucial to the system's functionality, as delays in data processing could lead to misclassification or failure to detect certain activities. Ensuring seamless operation of the data pipeline is necessary for the system to deliver timely and reliable classifications (Shi et al., 2016).

System-Level Testing

Real-World Activity Recognition

Users will be asked to perform a range of activities, such as walking, sitting, jumping, and climbing stairs, while wearing the device. This real-world testing will evaluate the system's ability to accurately classify activities under varying conditions. Performance metrics such as accuracy, precision, recall and F1-score will be calculated to measure the system's effectiveness in activity recognition. Additionally, the system will be tested to differentiate between similar activities, such as squats and lunges, as this is a common challenge in activity recognition (Lara & Labrador, 2013).

Power Consumption and Performance

The system's power efficiency will be tested over prolonged usage periods. As the device is wearable, its power consumption needs to be low to ensure long battery life without frequent recharging. Tests will measure energy usage during continuous operation and evaluate how different modes of activity impact power draw. Optimising power consumption is critical for wearable devices to maintain user convenience and long-term usage (Yurur et al., 2016).

Fault Tolerance and Error Handling

The system will be subjected to simulated errors, such as sensor disconnections, communication failures, and power fluctuations. These tests will assess the HAR system's fault tolerance and its ability to recover from errors without significant data loss. The system will be evaluated on its ability to detect, log, and recover from faults, ensuring it remains functional even in adverse conditions. Robust error handling is essential to ensure the HAR system can continue operating reliably in unpredictable environments (Palaniswami et al., 2013).

By conducting extensive component and system-level testing, this plan ensures that the HAR system is reliable, accurate, and robust for real-world fitness tracking applications. The evaluation focuses on ensuring real-time performance, reliability in diverse environments, and optimal power efficiency, all of which are essential for wearable technology.

6) Conclusion

In summary, this design proposal outlines the development of a machine learning-based Human Activity Recognition (HAR) system using an ADXL343 accelerometer integrated with a Raspberry Pi. The system is designed to classify specific physical activities—walking, sitting, jumping, and climbing stairs—by analysing acceleration data. By leveraging the interpretability and robustness of Decision Tree and Random Forest classifiers, our approach aims to provide accurate real-time activity recognition. The focus on feature extraction, including both time-domain and frequency-domain features, ensures that the models capture the nuances of different movements, enabling precise differentiation between activities.

Our system's emphasis on simplicity, efficiency, and effective error handling makes it a robust and practical solution for fitness tracking applications. Through a comprehensive evaluation strategy involving metrics such as accuracy, precision, recall, and cross-validation, the system is designed to ensure reliable performance in real-world scenarios. This HAR system offers significant value in the context of wearable technology, providing users with personalised insights into their physical activities. By focusing on targeted workout activities and excluding more complex functionalities, this prototype addresses a critical need for accessible and accurate fitness tracking solutions.

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8) Contribution Statement

This project was completed through the collaborative efforts of all team members. The specific tasks and responsibilities handled by each group member are as follows:

Nicholas Sloman: I was Responsible for outlining the Design Objectives and System Specifications. This included defining the overall goals of the (HAR) wearable device, identifying the key functionalities, hardware, and software components needed, and specifying the detailed requirements for the project.

Thorn Horwell:

I was responsible for developing the Machine Learning (ML) Model section and writing the Conclusion. My work involved selecting the appropriate ML models, outlining feature extraction methods, and summarising the project's key points and impact.

Paris Whiteley: I was responsible for writing the introduction and testing plan sections of this report. This included outlining the project scope, objectives and contextual relevance within the introduction and developing a testing plan to reinforce the system's accuracy and stability while aligning these sections with the overall design and system specifications.